

ConCaV: Confusion-triggered Captioning for Video Calls

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Abstract

Video calls simplify interpersonal exchanges, but communication partners quite frequently do not understand each other, especially when multiple users are connected. These situations are easily resolved in face-to-face encounters by paying attention to non-verbal cues, but go unnoticed in the online environment. In this paper, we therefore propose ConCaV, a video call system that recognizes confusion in facial expressions and in response activates audio captions. In a user-centered development process, we identified groups of action units that are associated with non-understanding. Preliminary evaluations show that non-understanding significantly drops when the dynamic captioning is applied.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interactive systems and tools**; *User models*; *Empirical studies in interaction design*; Collaborative interaction.

Keywords

Non-understanding; Confusion; Facial Expression; Action Unit; Captions; Conference; Video Call

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1 Introduction

Non-understanding is a common problem in video calls, and with multiple people connected, it is disruptive to the flow of the conversation if the speakers are constantly asked to repeat themselves. In this context, it is important to distinguish between non-understanding and misunderstanding [9]. In the case of non-understanding, listeners are aware of their comprehension deficit. Misunderstanding is – if at all – only retrospectively recognized by the interacting parties. It is typically caused by poor explanations of the speaker, whereas non-understanding may also have external causes. In particular, poorly received audio messages – a common problem in video calls – inevitably lead to non-understanding.

Multimodality mitigates this issue to some extent [37]. Particularly, captioning can resolve non-understanding of audio [17], as it provides an additional visual channel that complements speech. This can help fill in the gaps if speech fragments are lost due to background noise on the listener's side or a poor internet connection. Additionally, captions can help second language (L2) speakers to better understand spoken conversations [14, 19, 31, 38, 39]. This is especially relevant for online lectures or international project collaborations. On the other hand, if captions cause visual distraction, they can even hinder comprehension [21, 35]. As a result, it may not be desirable to display them all the time.

In this paper, we propose a video call system that activates audio captioning only when the user's facial expressions indicate non-understanding. **ConCaV** (**Confusion-triggered Captioning for Video calls**) uses real-time facial expression analysis to detect non-understanding and provide event-based dynamic captioning. While other cues could be used to trigger captioning, the advantage of facial expressions is that they can be obtained from webcam images and require no additional equipment or input from the user.

The design of the adaptive component is based on a user-centered development process. In a study with 39 subjects, we used a corrupted audio file to induce non-understanding. We then identified which facial expressions appear in webcam streams during the manipulated sequences. A preliminary evaluation identifies **ConCaV** as a promising tool for improving understanding in video calls.

2 Related work

CONCAV is based on two pillars. First, it must *detect non-understanding from facial expressions*. Non-understanding is a broad term that may encompass confused, expressionless, hesitant, or emotionless expressions [28]. As most of these expressions can be easily mistaken for a neutral state, research has mostly focused on detecting confusion. Emotion databases typically only contain annotated images for the basic emotions happy, anger, sadness, disgust, surprise, and fear [27] identified by Ekman and Friesen [12]. Since collecting training data from scratch is costly, alternative methods use the Facial Coding Action System (FACS) which defines action units (AUs) based on movements of the underlying muscle groups [13]. Table 1 summarizes studies that examine the relationship between confusion and AUs. Most frequently, AU4 and AU7 have been linked to confusion [7, 8, 11, 20, 22, 36]. However, it is the combinations of several AUs that characterize emotions. We therefore extracted AUs from participants of a video call and analyzed which combinations of AUs occurred most often.

The second pillar of **CONCAV** assumes that *captioning* can resolve non-understanding. Video captions provide a synchronized transcription of speech and sound for videos [33]. They were originally developed for people with hearing impairments [24], but also improve understanding of audio lectures [30, 39] and for L2 learners [10, 14, 31, 38]. Meeting tools like Zoom [26], Microsoft Teams [29], and Google Meet [18] offer live captioning, but the visual clutter can be distracting for users who do not need the captions and may even hinder understanding [21, 35]. Interactive captioning therefore transcribes only audio content that a user finds difficult to understand [32]. However, captioning only difficult words appears to be insufficient for understanding movies [16]. In order to provide sufficient captioning without unnecessary visual distraction, **CONCAV** captions all spoken discourse, but only after detecting non-understanding.

3 The CONCAV system: Confusion-triggered Captioning for Video calls

In this section, we present the user-centered design process that led to the development of the **CONCAV** (Confusion-triggered Captioning for Video calls) system. Its innovative feature is the dynamic captioning of a video call whenever the participant's expression, defined by facial action units, shows non-understanding. Given the ambiguity of other expressions related to non-understanding, we assume that the system primarily captures confusion. Hesitant or emotionless expressions, for instance, can also occur when a user is concentrated. In a video call, where participants tend to be highly focused, a clear distinction between

these states is not possible. In the following, we therefore use the terms 'non-understanding' and 'confusion' interchangeably. The system consists of three components:

UserInterface. Via a session link, participants can open a standard meeting interface with a minimal set of control buttons in their browser. A special host interface provides additional buttons to start and manage the session. Video streams of participants and the host are displayed in a tiled window.

ExpressionAnalyzer. AUs are extracted from webcam frames in real-time at a rate of 7 frames per second. We used the OpenFace [25] toolkit to extract binary values (1 : present, 0 : absent) for 18 facial AUs [3, 4].

SessionHandler. An HTTP server manages the communication between the participants and runs the *adaptation module*. It receives a continuous stream of webcam images and sends individual frames to the ExpressionAnalyzer for the extraction of facial AUs. Algorithm 1 determines whether the extracted AUs express confusion. If confusion occurs due to interruptions like momentary breaks in the user's internet connection or sudden background noise (e.g., a passing car), it is considered a fleeting state. In these situations, we expect that future audio transmissions will be clear once again. Consequently, captioning future audio output proactively would not benefit the user. Instead, proactive captioning should only be applied when confusion is expected to persist. **CONCAV** therefore only triggers captioning when confusion last for one second or longer. At an extraction rate of 7 frames per second, this means that an alert is triggered if the AUs of at least 7 consecutive frames indicate confusion.

3.1 User-centered development of the adaptation logic

To develop a mechanism that effectively detects and responds to confusion, we conducted a study with 39 participants. They were invited to test **CONCAV** remotely using their own devices.

Participants. Using convenience sampling, we recruited 39 subjects of diverse age and cultural background (26 male, 13 female, mean age = 28.8 ± 13.3). All were native ($n = 27$) or advanced L2 speakers (C1: $n = 7$, B2: $n = 3$, B1: $n = 2$). Each participant was sent an invitation for a one-to-one meeting. Written informed consent was obtained for the recording and analysis of the session. The participants volunteered without expectations of monetary or equivalent compensation.

Procedure. After entering demographic data, the participants were instructed by the host to keep their camera on for the session. The host explained that they would now watch a documentary and

Table 1: Action units that have been associated with confusion.

Source	Task(s) of the study	Action units
Yasser et al. [36]	Answering personal questions	AU4, AU7, AU12
Kaushik et al. [22]	Impossible scheduling/logical reasoning	AU4, AU7, AU5, AU12/AU14 (smirk), AU15
D'Mello et al. [11]	Multimedia e-learning	AU4, AU7, (AU12)
Borges et al. [8]	Misleading navigation prompts	AU4, AU15, AU25, AU26, AU27
Heck et al. [20]	TV news reports	AU4, AU6, AU7, AU10, AU17, AU23
Bishay et al. [7], [1]	Confusing advertisements	AU1, AU4, AU7, AU9, AU14, AU15, AU18, AU26

should answer some questions about its content afterwards. The pre-recorded video then started playing in full-screen mode.

In addition to real-time extraction and analysis, the time-stamped AUs were logged to a CSV file for offline analysis. In the experiment, dynamic captioning was enabled for half of the participants using a dummy algorithm. The algorithm defined confusion as an increased activation of one of the two AU sets $AU[4, 7, 10, 17]$ or $AU[4, 7, 25, 26]$. The AU sets were chosen based on acted displays of confusion from 3 subjects of a small pilot study. Once activated, the captions persisted until the next confusion trigger in the video was reached. At the end of the video, 12 questions about its content were displayed. In a subsequent second playback of the video, the participants performed a self-assessment by pressing the space bar whenever they recalled having felt confused during the first replay.

Video material. Confusion was generated by a pre-recorded 10-minute video about the life of Julius Caesar. Several sequences were manipulated to simulate poor connection or unclear pronunciation of the speaker, which are common causes for confusion in video calls: (a) audio-visual disparity (e.g., narration about Caesar while showing a picture of Napoleon), (b) background noise, (c) mumbling, (d) fast speech, (e) fragmented speech.

3.2 Data-driven specification of the confusion detection algorithm

To assess the validity of our confusion triggers, we verified whether their display coincided with self-assessed confusion. For most participants, mumbling (71.8%) and fragmented speech (64.1%) caused confusion. Audio-visual disparity (17.9%), background noise (37.2%), and fast speech (34.2%) only confused a subset of participants. Consequently, we only labeled triggering sequences as ‘confusing’ for a participant if the self-assessment also identified them as such. In the following, we refer to these sequences as ‘trigger periods’.

For the experiment, we used a dummy algorithm to identify confusion. It is important to note that the implemented AUs were based on a small pilot study with 3 subjects. We therefore proceeded with a more comprehensive analysis of the AUs using the data of our 39 participants. We first defined *confusion AUs* as AUs that were present in trigger periods. Figure 1 shows that a large part of the recorded frames contained AU4, AU5, AU6, AU7, AU10, AU12, and AU14.

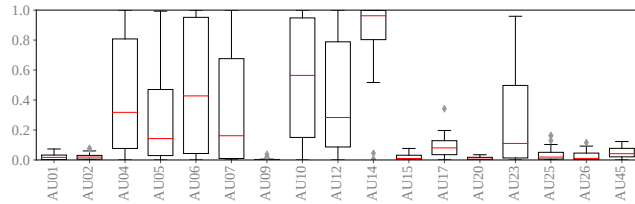


Figure 1: Fraction of frames $\in [0,1]$ from trigger periods in which AUs were present across participants.

However, an individual AU can be associated with multiple emotions [5, 15, 23]. Specific combinations, on the other hand, can help to accurately identify an emotion [5, 23]. Therefore, we applied the *a priori* algorithm [2] to associate entire sets of AUs with confusion.

The *a priori* algorithm confirmed that AU4, AU5, AU6, AU7, AU10, AU12, or AU14 in isolation correlate with confusion. This association tends to persist when other confusion AUs co-occur, but is lost when another AU that is unrelated to confusion is present. Table 2 summarizes the results.

Table 2: Combinations of facial action units associated with confusion as identified by the *a priori* algorithm [2].

AU	Description	Co-occurring AU sets associated with confusion
AU4	Brow Lowerer	{6}, {10}, {14}, {6,10}
AU5	Upper Lid Raiser	{6}, {10}, {14}, {6,10}, {6,14}, {10,14}, {6,10,14}
AU6	Cheek Raiser	{4}, {5}, {10}, {14}, {4,10}, {5,10}, {5,14}, {10,14}, {5,10,14}
AU7	Lid Tightener	
AU10	Upper Lip Raiser	{4}, {5}, {6}, {14}, {4,6}, {5,6}, {5,14}, {6,14}, {5,6,14}
AU12	Lip Corner Puller	{14}
AU14	Dimpler	{4}, {5}, {6}, {12}, {14}, {5,6}, {5,10}, {6,10}, {5,6,10}

Algorithm 1 then checks for these AU sets (CONFUSIONSETS) in the output from the ExpressionAnalyzer. Since captioning is only triggered if confusion persists for at least 7 frames, CONCAV counts the number of consecutive frames from which AU sets were extracted that indicate confusion. We call such a period CONFUSION-INTERVAL.

Algorithm 1: Confusion Detection

```

Input: Stream  $\mathcal{A} = \langle auList_1, auList_2, \dots, auList_n \rangle$ 
1 function detectConfusion( $auList$ )
2    $interruption \leftarrow 0$ ;  $confusionInterval \leftarrow 0$ 
3   forall  $auList \in \mathcal{A}$  do
4     if  $auList \in confusionSets$  then
5        $confusionInterval++$ 
6        $interruption \leftarrow 0$ 
7     else
8        $interruption++$ 
9       if  $interruption \leq \tau$  then
10         $confusionInterval++$ 
11      else
12         $confusionInterval \leftarrow 0$ 
13  if  $confusionInterval \geq 7$  then
14    triggerCaptioning()
15  return
```

In order to account for possible inaccuracies in the OpenFace software, frames without confusion only terminate the interval if the duration of the INTERRUPTION exceeds sensitivity threshold τ . Based on our observations of the typical duration of an interruption, we set $\tau = 2$. Increasing the sensitivity threshold would have little

Table 3: AU sets that have been empirically linked to basic emotions. AUs that we associate with confusion are highlighted.

Action units						Source
Happiness	Sadness	Fear	Anger	Surprise	Disgust	
{12,25}	{4,15}	{1,4,20,25}	{4,7,24}	{1,2,25,26}	{9,10,17}	Benitez-Quiroz et al. [5]
{1,2},{6,12},{7,26}	{4,7,20}	{1,2},{5,26}	{4,7}			Kohler et al. [23]
{6,12}	{1,4,15}	{1,2,4,5,7,20,26}	{4,5,7,23}	{1,2,5,26}	{9,15,16}	Farnsworth [15]

effect on reducing the number of unintended gaps (slope < 1), but it would substantially increase the risk of false confusion alerts.

3.3 Preliminary system evaluation

The dummy algorithm of the test system was found to perform barely better than chance. The two AU sets that the dummy algorithm used to detect confusion did not sufficiently cover the expressions of confusion in the larger pool of participants (cf. Table 2). We suspect that the limited amount of data from only 3 subjects and the reliance on acted expressions of confusion are responsible for the poor performance. While not initially intended, this allowed us to fairly evenly split the recorded samples into a treatment group (with captioning activated) and a control group. Samples were labeled ‘captioned’ when the dummy algorithm had activated captioning, and ‘uncaptioned’ otherwise.

Using a between-subjects design, we conducted a preliminary assessment of dynamic captioning. For each trigger period, we extracted the number of frames with confusion as a fraction of the total recorded frames. A two-sample t-test for independent samples was used to determine whether captioned sequences reduced the number of frames with confusion. Test statistics were calculated using Welch’s t-test to account for unequal variances and confirmed that captioned sequences produced significantly less frames with confusion ($t = 2.907$, $p\text{-value} = .004$). Participants expressed confusion in 43.82% of the frames from uncaptioned sequences, compared to only 27.98% in captioned sequences.

4 Discussion and conclusion

CONCAV addresses issues of non-understanding in video calls by detecting when a user is confused and proactively captioning future audio transmissions. Emotion recognition based on single AUs is unreliable. Table 3 shows that AU4 and AU7, which have been empirically linked to confusion (see Table 1), are also present in expressions of sadness, fear, and anger. AU6 or AU12 alone indicate confusion, but in combination expresses happiness [15, 23].

By recognizing confusion only when very specific AUs co-occur, **CONCAV** substantially reduces to risk of false alerts. Although small displays of confusion may go unnoticed, constant monitoring of the user’s state ensures that persistent non-understanding is reliably detected. We will use the insights presented in this paper to extend our experimental investigations and address the remaining technical limitations of **CONCAV** as discussed in the following.

Confusion detection. The expression of confusion can vary across individuals [34]. Some display expressions similar to amusement, while others show features resembling irritation. Therefore, it could

be beneficial to collect training data during the video call to define user-specific confusion sets. Additional research is needed to determine whether the thus improved accuracy justifies the inconvenience of the additional data collection caused to users.

Caption generation. In our experiment, we used manually generated captions. If these are replaced by auto-generated captions, the accuracy of the transcript is heavily reliant on the clarity of the spoken input [6]. For best results, captions should be generated on the speaker’s end. This can compensate for lost speech due to poor connection or noise at the receiver’s end. However, issues related to a low-quality microphone or noise on the speaker’s end still persist. Our next steps will therefore be to implement auto-generated captions and test **CONCAV** under natural distractions.

Captioned sequences. Captioning should only be active when a user is confused. If the visual distraction is hindering understanding, captions should be turned off. The current version of **CONCAV** does not support this mechanism. However, it can be implemented by continuing to monitor the user’s state after captioning has been activated. If captions cause distraction, we expect this to be reflected in persistent confusion. One challenge we may encounter is that most of the AUs associated with confusion are located in the eye region and may be affected by squinting while reading the captions. If such an effect exists, it must be accounted for in the design of the caption deactivation mechanism.

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